High-Fidelity, Multiqubit Generalized Measurements with Dynamic Circuits

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Generalized measurements, also called positive operator-valued measures (POVMs), can offer advantages over projective measurements in various quantum information tasks. Here, we realize a generalized measurement of one and two superconducting qubits with high fidelity and in a single experimental setting. To do so, we propose a hybrid method, the "Naimark-terminated binary tree," based on a hybridization of Naimark's dilation and binary tree techniques that leverages emerging hardware capabilities for midcircuit measurements and feed-forward control. Furthermore, we showcase a highly effective use of approximate compiling to enhance POVM fidelity in noisy conditions. We argue that our hybrid method scales better toward larger system sizes than its constituent methods and demonstrate its advantage by performing detector tomography of symmetric, informationally complete POVM (SIC POVM). Detector fidelity is further improved through a composite error-mitigation strategy that incorporates twirling and a newly devised conditional readout error mitigation. Looking forward, we expect improvements in approximate compilation and hardware noise for dynamic circuits to enable generalized measurements of larger multiqubit POVMs on superconducting qubits.

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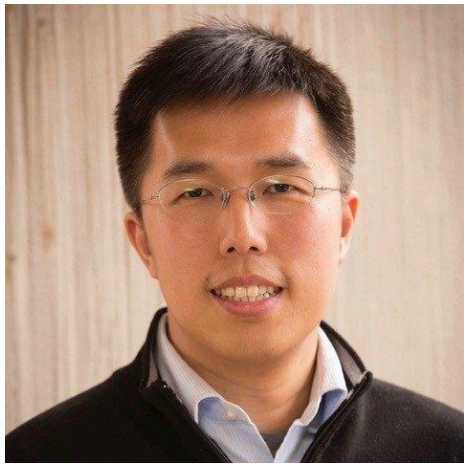

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Background

PVM is a set $\{\Pi_i\}$ of positive operators s.t.

- (Orthogonality) $\Pi_i \Pi_j = \delta_{ij} \Pi_i$
- (Completeness) $\sum_{i=1}^{M} \Pi_i = I$
- (Born rule) P(i) = $\mathrm{Tr}(\rho \Pi_i)$

POVM is a set $\{F_i\}$ of positive operators s.t.

- (Completeness) $\sum_{i=1}^{M} F_i = I$
- (Born rule) P(i) = $\mathrm{Tr}(\rho F_i)$

A POVM can have more elements than a PVM

Notation:

- **M**: number of POVM elements
- **n**: number of qubits
- **d** = $2^{\mathbf{n}}$: dimension of the system

Applications

IC POVM: at least d^2 linearly independent operators

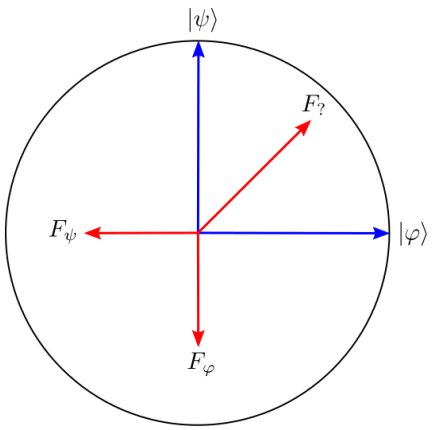
Observable estimation

$O = \sum_{i=1}^{M} c_i F_i$

$\implies \langle O \rangle = \mathrm{Tr}(\rho O) = \sum_{i=1}^{M} c_i \mathrm{Tr}(\rho F_i) = \sum_{i=1}^{M} c_i P(i)$

with $\quad c_i = \mathrm{Tr}(D_i O) \quad$ for a suitable dual basis $\{D_i\}$

Unambiguous state discrimination



$$F_\psi = \frac{1}{1+|\langle\varphi|\psi\rangle|}|\varphi^\perp\rangle\langle\varphi^\perp|$$

$$F_\varphi = \frac{1}{1+|\langle\varphi|\psi\rangle|}|\psi^\perp\rangle\langle\psi^\perp|$$

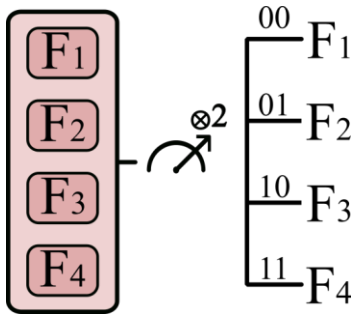
$$F_? = \mathrm{I} - F_\psi - F_\varphi.$$

Naimark's dilation

For rank-1 POVM elements $F_i = |\psi_i\rangle\langle\psi_i|$:

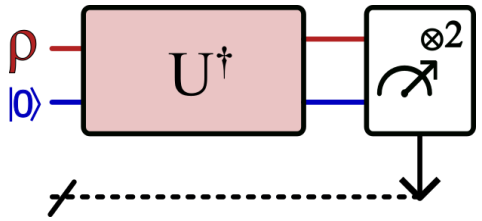
1. Arrange $|\psi_i\rangle$ column vectors to form an $d \times M$.
2. Extended to a $M \times M$ unitary matrix:

$$U = \begin{pmatrix} \psi_{11} & .. & \psi_{M1} \\ \vdots & & \vdots \\ \psi_{1d} & .. & \psi_{Md} \\ * & .. & * \end{pmatrix} \in \mathbb{U}(M)$$



3. Prepare ancilla in $|0\rangle^{\otimes n_A}$ and apply $U^\dagger$ to the joint system.
4. Measure in the computation basis:

$$\mathrm{P(i)} = \mathrm{Tr}(|\psi_i^{\mathrm{ext}}\rangle\langle\psi_i^{\mathrm{ext}}|(\rho_A \otimes \rho)) = \mathrm{Tr}(|\psi_i\rangle\langle\psi_i|\rho) = \mathrm{Tr}(F_i\rho).$$



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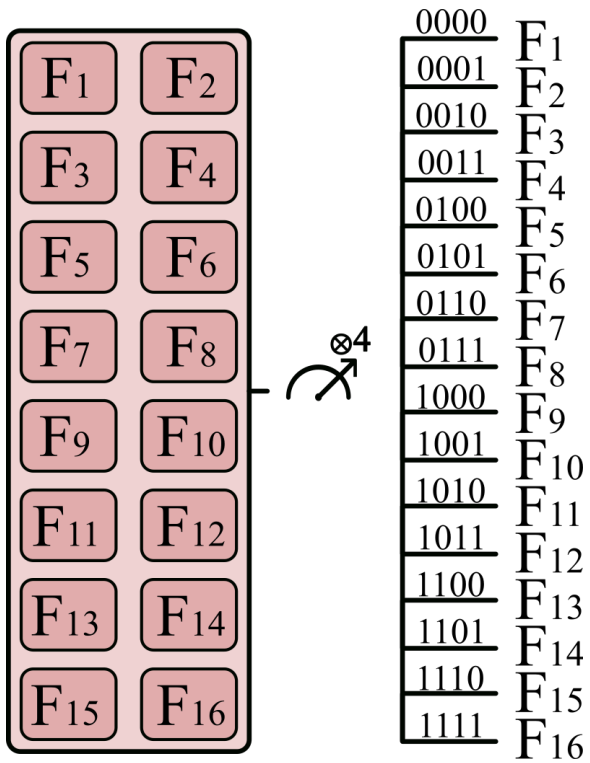
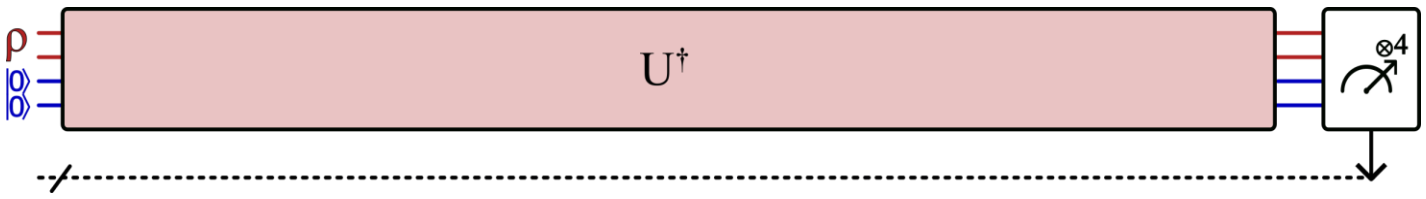




Not scalable

Binary search

1. Split the original POVM into two sets:

$$B_0 := \sum_{i=1}^{M/2} F_i \quad \text{and} \quad B_1 := \sum_{i=M/2+1}^{M} F_i. \tag{2}$$

2. Measure a two-outcome POVM via Stinespring dilation:

$$U^B = \begin{pmatrix} A_0 & * \\ A_1 & * \end{pmatrix} \in \mathbb{U}(2d). \tag{3}$$

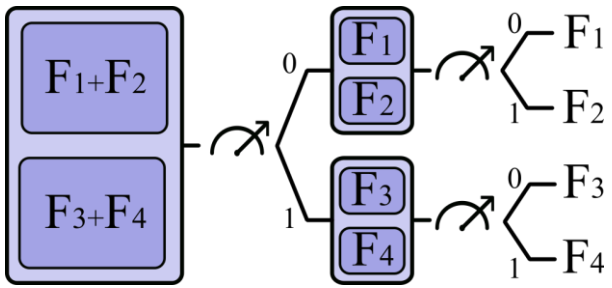
   where $A_0 = \sqrt{B_0}$ and $A_1 = \sqrt{B_1}$.

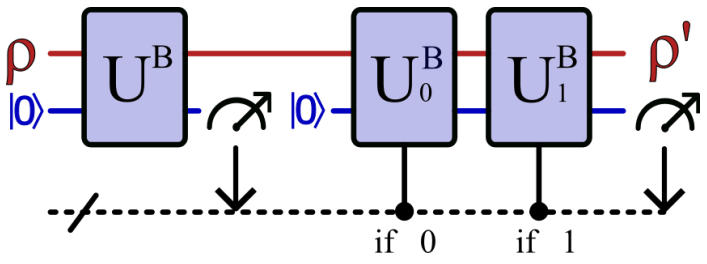
3. The post-measurement state is disturbed: $\rho = \frac{A_0 \rho_0 A_0^\dagger}{\mathrm{Tr}(A_0 \rho A_0^\dagger)}$

4. Account for the post-measurement state by modifying the Kraus operators:

$$A_1 = K A_0^{-1} \tag{4}$$

   where $K = \sqrt{\sum_{i=a}^{b} F_i}$.





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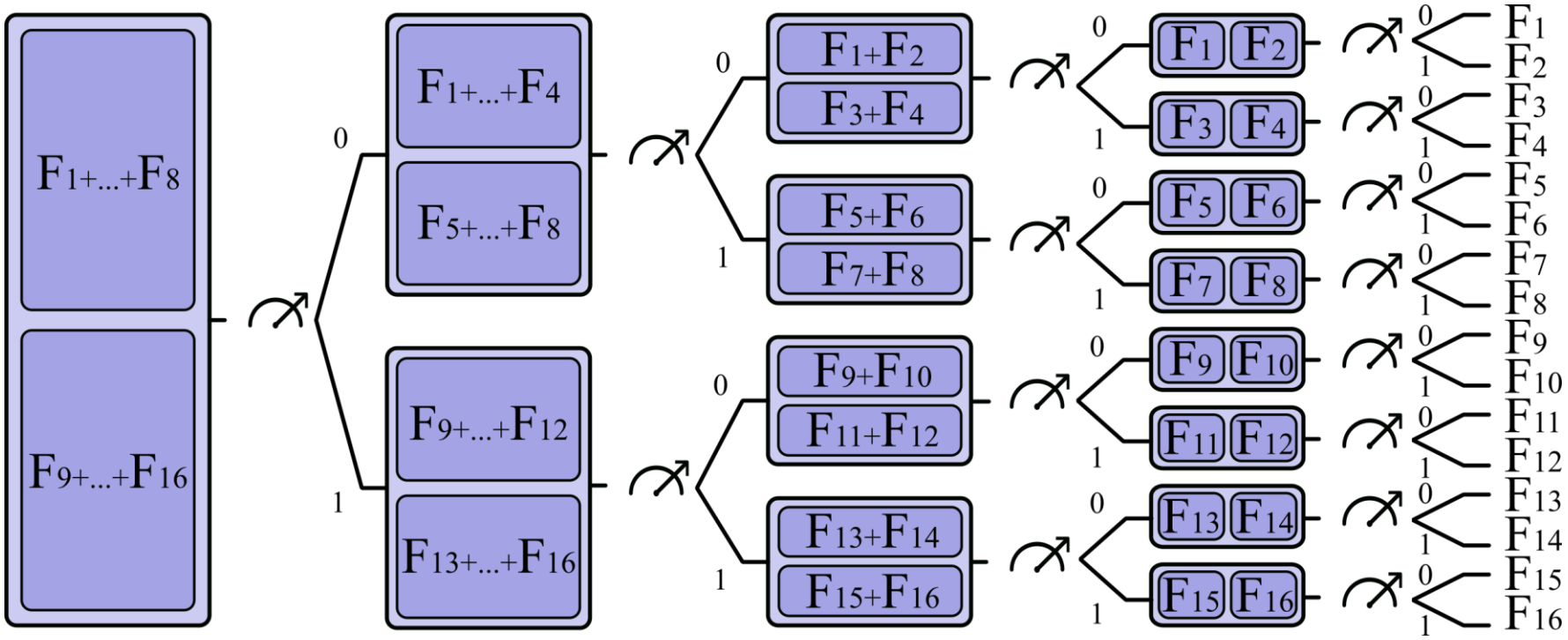
F1+...+F8
F9+...+F16
F1+...+F4
F5+...+F8
F9+...+F12
F13+...+F16
F1+F2
F3+F4
F5+F6
F7+F8
F9+F10
F11+F12
F13+F14
F15+F16
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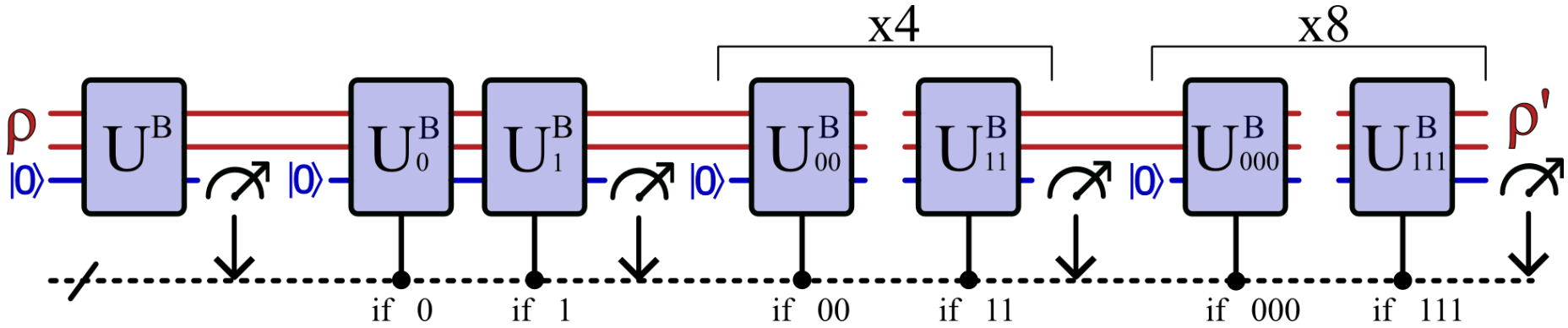
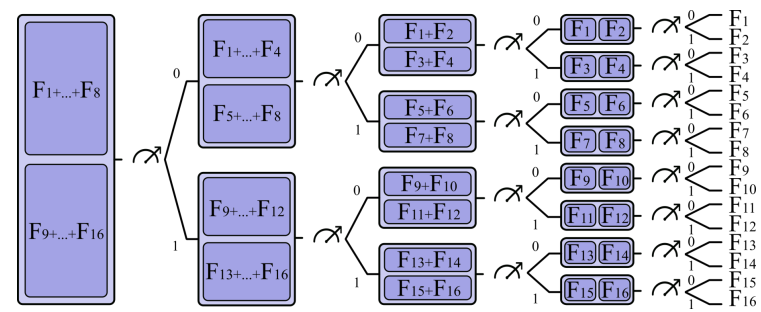
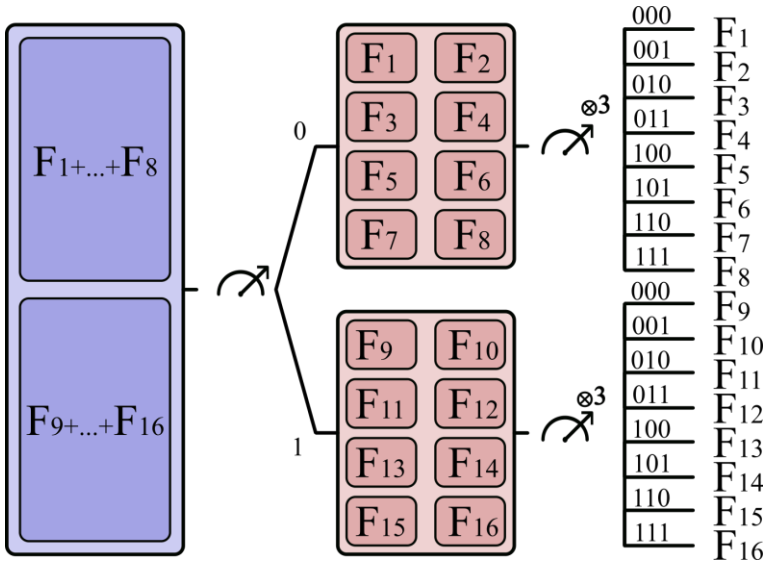


1. Unutilized resource $\implies$ optimization potential
2. Noisy mid-circuit measurements and feed-forward operations



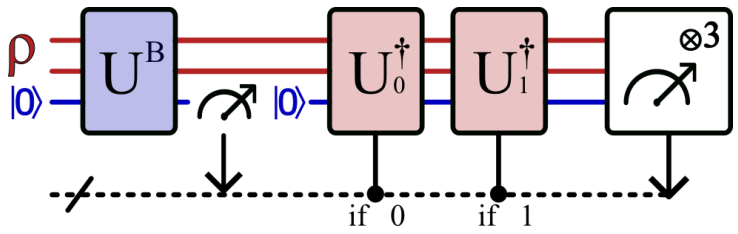
Hybrid approach



1. Apply $m = \log_2\left(\frac{M}{2d}\right)$ binary search steps.
2. Interrupt the search and modify the vectors $|\tilde{\psi}_i\rangle = K_{b^{(m)}}^{-1\dagger}|\psi_i\rangle$.
3. The new POVM defined by $\tilde{F}_i = |\tilde{\psi}_i\rangle\langle\tilde{\psi}_i|$ satisfies the completeness:

$$\sum_{i=a}^{b} |\tilde{\psi}_i\rangle\langle\tilde{\psi}_i| = K_{b^{(m)}}^{-1\dagger}\left(\sum_{i=a}^{b} F_i\right)K_{b^{(m)}}^{-1} = I,$$

4. Measure $\{\tilde{F}_i\}$ via (conditional) Naimark's dilation.



Experiments on hardware

One- and two-qubit SIC-POVMs:

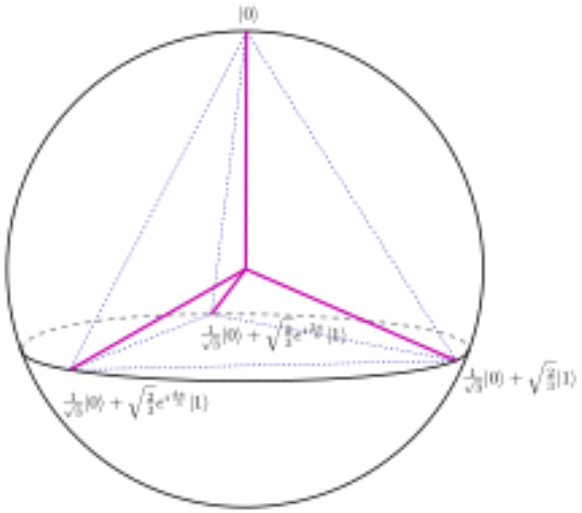
1. Span the space of Hermitian operators
2. Symmetric property

$$\mathrm{Tr}(F_i F_j) = \frac{\delta_{ij} + 1}{d + 1}$$

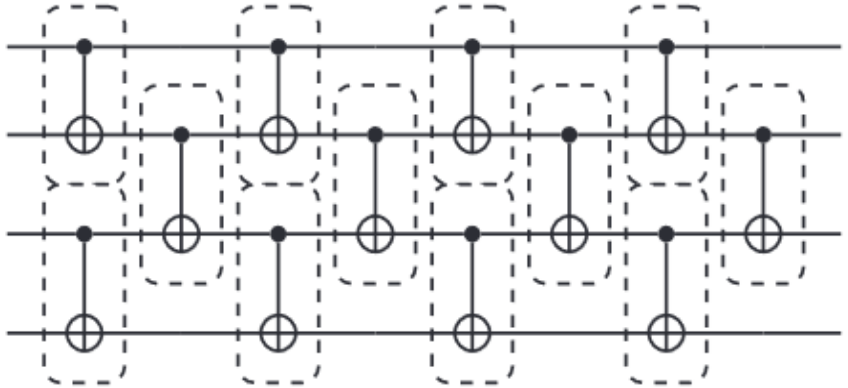
3. Optimal for tomography

Detector tomography:

1. Known initial states, unknown detector (POVM)



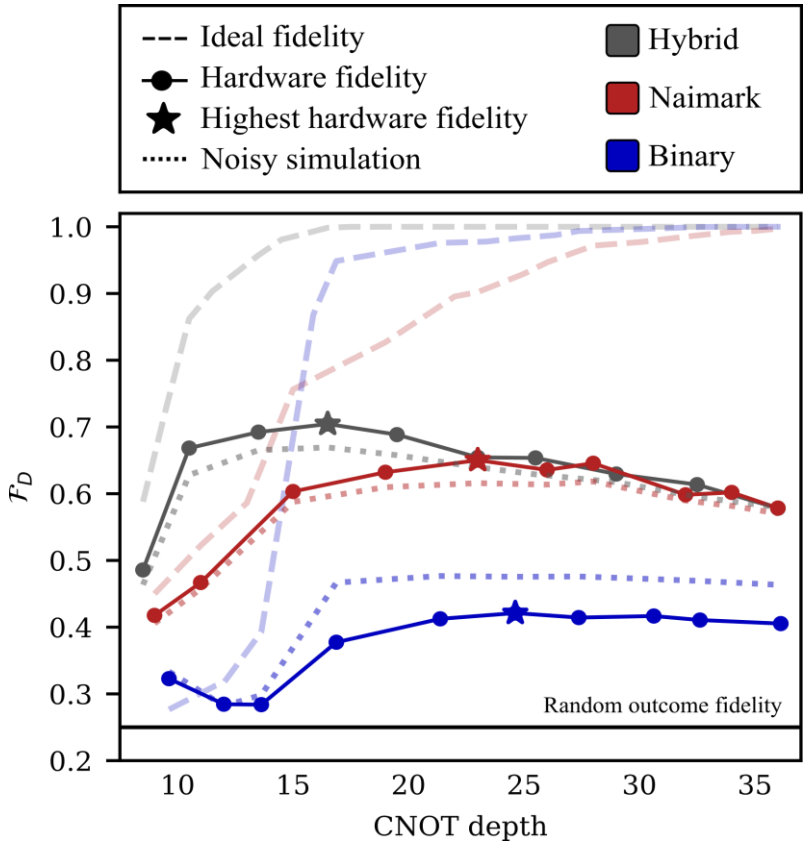
Approximate compiling



$$\Delta(U_T, U_C) = 1 - \frac{|Tr(U_T^\dagger U_C)|}{2^n}$$

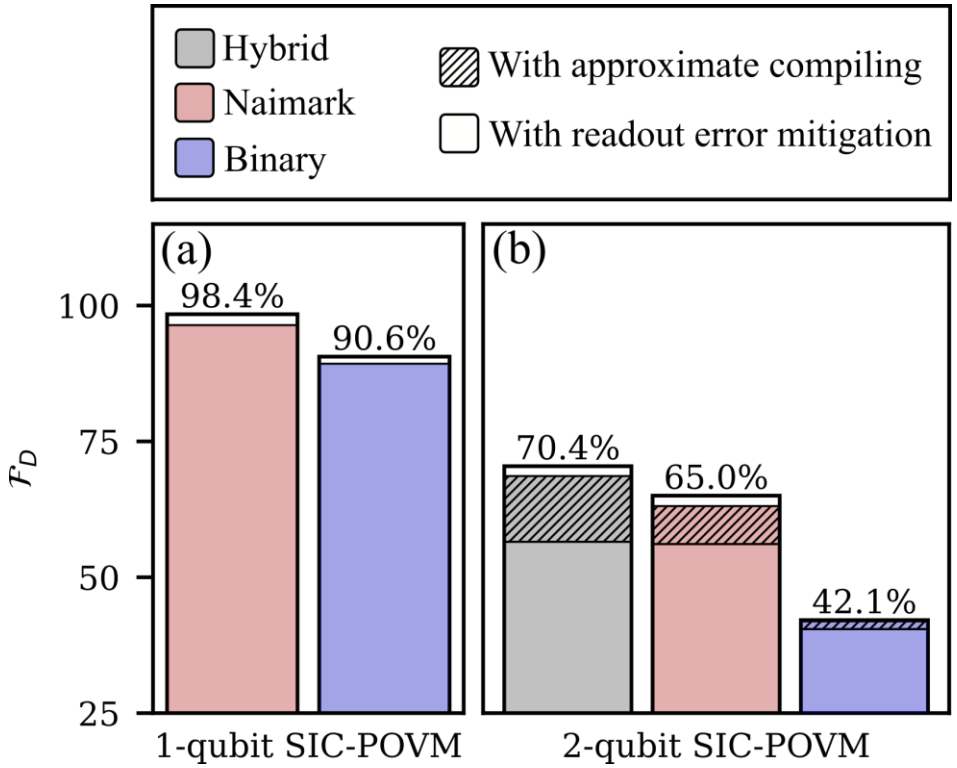
Madden and A. Simonetto, ACM Transactions on Quantum Computing **3**,1 (2022)

Approximate compiling



1. Need to decompose unitaries in native gates.
2. Exact compiling techniques are infeasible.
3. $\implies$ Compile approximately and find an optimal tradeoff between approximation error and circuit noise.
4. Close agreement with a depolarizing noise model.

Hardware results

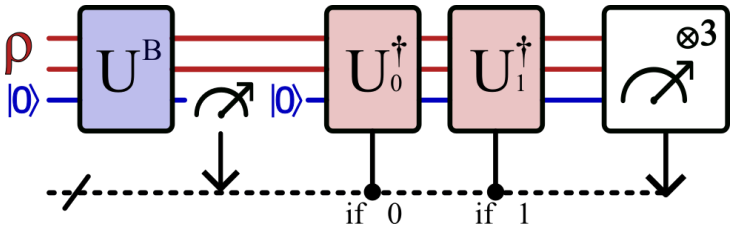


1. Hybrid outperforms Naimark and binary.
2. Binary dominated by mid-circuit noise.
3. Higher performance differential is expected for larger systems.

Example for a minimal IC-POVM

M	Naimark	Binary	Hybrid
$M = 4^n$	16^n	$2n \cdot 4^{n+1}$	$n \cdot 4^{n+1}$

TABLE I. Upper bounds on the number of CNOTs required for POVM implementation.

Outlook

1. **Scaling to more qubits:** faster feed-forward, error suppression techniques
2. **Products of local POVMs:** cover more qubits on the chip
3. **Parametric POVMs:** hardware efficient ansaetze